

- List the common parasites which affect the eye (Sparganosis, Cysticercosis, and Hydatid disease, *Visceral larva migrans as Toxocara canis, Toxocara cati, Trichinella spiralis, Onchocerca volvulus, Loa loa, Acanthamoeba, Toxoplasma gondii, Trypanosoma cruzi, ocular myiasis, phithirus pubis*).
- Demonstrate knowing the life cycle (definitive host, intermediate host, habitat) of the selected parasites (*Onchocerca volvulus, Loa loa, Acanthamoeba*).
- Identify the sources of infection, the infective and diagnostic stages of the selected parasites (*Onchocerca volvulus, Loa loa, Acanthamoeba*).
- Describe disease pathogenesis, and main clinical presentations of the selected parasites (*Onchocerca volvulus, Loa loa, Acanthamoeba*).
- Describe laboratory diagnosis, imaging and pathological studies of lesions caused by these selected parasites (*Onchocerca volvulus, Loa loa, Acanthamoeba*).
- Outline the general principle of treatment, prevention and control of the selected parasites (*Onchocerca volvulus, Loa loa, Acanthamoeba*).

Common parasites affecting the eye	▪ Sparganosis ▪ Cysticercosis ▪ Hydatid disease ▪ <i>Visceral larva migrans as Toxocara canis, Toxocara cati</i> ▪ <i>Trichinella spiralis</i> ▪ <i>Onchocerca volvulus</i> ▪ <i>Loa loa</i> ▪ <i>Acanthamoeba</i> ▪ <i>Toxoplasma gondii</i> ▪ <i>Trypanosoma cruzi</i> ▪ <i>Ocular myiasis</i> ▪ <i>Phithirus pubis</i>
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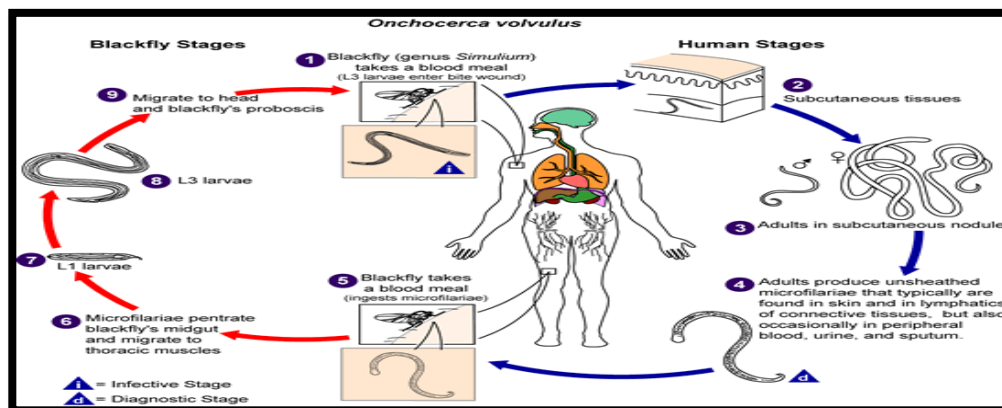
Onchocerca volvulus

Disease	Onchocerciasis, cutaneous filariasis Sudanese or river-blindness
Geographical distribution	Tropical and South Africa, Central America, Yemen and Saudi Arabia
Definitive host (D.H.)	Man
Intermediate host (I.H.)	<i>Simulium</i> (black fly), Cyclo-developmental transmission (the organism undergoes

	developmental changes with no multiplication).
Habitat	Adult worms and microfilariae are located in nodules in subcutaneous tissues
Infective stage (I.S.)	Infective 3 rd stage filariform larva
Diagnostic stage (D.S.)	Microfilariae
Mode of infection	Filariform larvae are deposited on human skin when infected fly takes blood meal & penetrate the skin through bite wound

Life cycle

1. The adult worms live in the subcutaneous connective tissues and lymphatic spaces, (at least one male and one female). As the infection progresses these worms become encapsulated in fibrous nodules. The adult worms live up to 15 years or more in their host.
2. Viviparous females produce many unsheathed microfilariae which migrate out of the nodule to enter the dermis of the skin and the connective tissue.
3. The microfilariae also migrate to the eye and other organs of the body, but generally do not enter the circulation.
4. *Simulium* (black fly) ingests the microfilariae from the skin of infected persons with its blood meal.
5. Inside the black fly, development of microfilariae into infective larvae as well as mechanism of transmission resembles that of *W. bancrofti* in its mosquito host.
6. In subcutaneous tissue, development of larvae into adult male and female worms taking several months.



Pathogenesis and clinical picture:

- Adults lie coiled in subcutaneous fibrous nodules (Onchocercoma) & do not cause any other symptoms.
- The main pathogenic effect is from microfilariae. Reaction around them causes tissue damage, loss of skin elasticity and infiltration by inflammatory cells. This occurs in skin and in the eyes.

1. Onchocerca nodule (Onchocercoma): It is painless, firm, movable subcutaneous nodule, variable in size. Mainly found on bony parts (scalp, elbow, knee, iliac crest, scapula and ribs). Inside lesion, adult worms and microfilariae are coiled in masses & are surrounded by tissue reaction from the host.

2. Skin lesions:

- a. In Latin Americans: Inflammatory skin reaction in the upper part of the body, causes skin thickening & dark coloration. Loss of elasticity causes premature senility or leonine faces.
- b. In Africans: It affects back, lower trunk and lower limbs. There is skin wrinkling, loss of elasticity, thickening of skin & areas of depigmentation (leopard skin). Loss of elasticity in lower abdomen causes skin folding simulating inguinal hernia (hanging groin).
- c. In Yemen area: Skin manifestations are localized mainly in lower limbs & may be in one limb, arm or trunk as dark discoloration (sowda), with inflammation, intense itching, swelling & enlargement of regional lymph nodes.

3. Ocular lesions (River or Sudan blindness): It is usually called River-blindness as it occurs in people living near fast running streams where *Simulium* breeds.

- Ocular lesions are due to:

- Toxic or allergic reactions to living and dead microfilariae.
- Migration of microfilariae from the nodules into the eyes.
- Lesions: Keratitis, iridocyclitis, optic nerve atrophy (complete blindness).
- Manifestations: Photophobia, lacrimation, blurring of vision, foreign body sensation up to blindness.

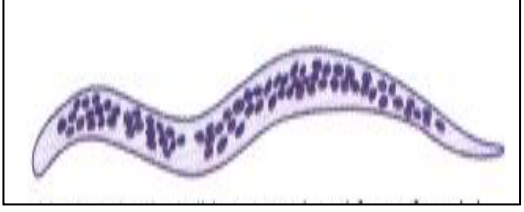
Diagnosis:

I. Clinical diagnosis: History in residence in endemic area+ clinical picture.

II. Laboratory diagnosis:

1. Direct methods:

- Detection of microfilariae in aspirate of nodules or bloodless skin snips.
- Biopsy of nodules for detection of adults surrounded with fibrous tissue.

Microfilariae - 300 μm $\times$ 9 μm - None sheathed. - Head end is slightly enlarged. - Long pointed tail with no nuclei. - No peridocity.	
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2. Indirect methods (Immunodiagnosis):

- Serological test: Detect specific antibodies.
- DEC patch test: Apply a small amount of 10 % 100 mg Diethyl Carbamazine (DEC) in Nivea Lanolin cream to an area of skin (5mm) and cover the area with a dressing. A positive test shown by a papular eruption developing 8-24 hours after applying the DEC. The reaction is an allergic response to dead microfilariae at the site of application.

Treatment:

- Nodulectomy: Removal of the nodules especially from the head reduces the risk of ocular lesions.
- Anti-filarial drugs: Ivermectin, DEC (Hetrazan).

Prevention and control:

- Treatment of the patients.
- Vector control: Control of *Simulium* fly is difficult as larvae and pupae attach to submerging rocks in river.
- Personal protection by repellent, screening and insecticides.

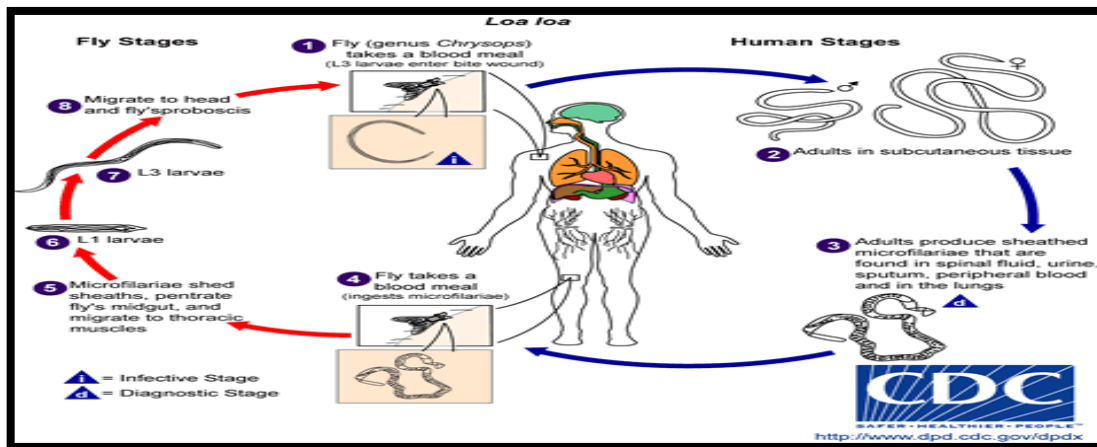
Loa loa (African eye worm)

Disease	Loiasis, 'calabar swellings' or "fugitive swellings"
Geographic distribution	West and Central parts of Tropical Africa
Definitive host (D.H.)	Man
Intermediate host (I.H.)	<i>Chrysops</i> fly, Cyclo-developmental transmission (the organism undergoes

	developmental changes with no multiplication).
Habitat	Adult worms live in subcutaneous tissues and can be seen under conjunctiva of the eye. Sheathed microfilariae are present in the blood in day time
Infective stage (I.S.)	Infective 3 rd stage filariform larva
Diagnostic stage (D.S.)	Microfilariae
Mode of infection	Filariform larvae are deposited on human skin when infected fly takes blood meal & penetrate the skin through bite wound

Life cycle:

1. Adults live free in the subcutaneous tissue of man, migrate and can be seen under conjunctiva of the eye. They may go from one eye to the other across the bridge of the nose.
2. The sheathed microfilariae appear in peripheral blood only during the day, taken by female *Chrysops* when sucking blood and develop into infective filariform larvae.



Pathogenesis and clinical picture:

- The adult worms have tendency to wander through the subcutaneous connective tissues, provoking temporary inflammatory reactions.
- Localized allergic reactions to the parasite antigen called Calabar swelling is characteristic of the disease.

1. Calabar swelling (fugitive swelling):

- Painless swelling, usually seen on the extremities (hand, forearm, elbow, ankle) but may appear anywhere on the body, itchy, non- pitting, about the size of hen's egg that appear suddenly & disappear within 2-3 days & reappear again (fugitive).

- They are allergic in nature & never contain microfilariae or worms.

2. Ocular manifestations: Adult worms moving under conjunctiva can cause irritation, itching, lacrimation, edema and painful swelling. They appear and disappear within 10-15 minutes usually leaving no traces. *Loa loa* does not cause blindness.

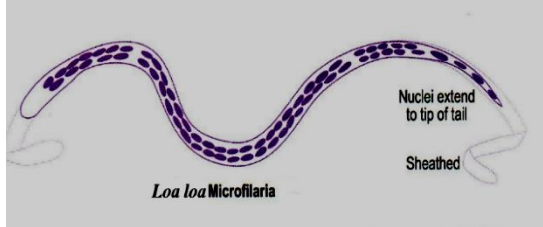
3. Complications: May occur when microfilariae invade vital organs e.g. CNS producing meningo-encephalopathy.

Diagnosis:

I. Clinical diagnosis: History (residence in endemic area), clinical picture (calabar swelling and detection of adult worms under conjunctiva).

II. Laboratory diagnosis:

1. Direct method: By detection of the characteristic microfilariae in blood film prepared during the daytime.

Microfilaria of <i>Loa loa</i> - 250 × 8μm, with tightly sheath and angular curves. - Nuclei reach the tail end. - Diurnal periodicity (between 8 am-5 pm).	 The diagram illustrates a single adult <i>Loa loa</i> microfilaria. It is a long, slender, wavy worm with a distinct head at the left and a tail at the right. The worm is covered by a thin, wavy line representing its sheath. Inside the body, numerous dark, oval-shaped nuclei are visible, extending from the head towards the tail. Labels include 'Loa loa Microfilaria' at the bottom left, 'Nuclei extend to tip of tail' at the top right, and 'Sheathed' at the bottom right.
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2. Indirect methods:

a. Serological tests: ELISA, CFT, IHAT, IFAT.

b. Blood examination: Eosinophilia.

c. Molecular diagnosis: PCR.

Treatment:

1. Surgical removal of the adults if seen under conjunctiva

2. Anti-filarial drugs: DEC (Hetrazan), Ivermectin.

Prevention and control:

1. Treatment of patients.

2. Control of *Chrysops*.

Acanthamoeba keratitis

Acanthamoeba keratitis is a chronic, progressive, ulcerative disease of the eye. It occurs in healthy individuals particularly those using contact lenses.

Mode of transmission:

1. Direct invasion of the amoebae through minor trauma in the cornea caused by the use of soft contact lenses.
2. Also, the amoebae enter the eye by the use of contaminated contact lens, cleansing solution or swimming in contaminated water.

Pathogenesis and clinical picture:

1. Ulcers of the cornea are painful and resistant to treatment with usual antifungal, antibacterial and antiviral agents. If it is not successfully treated, the disease progresses to corneal perforation, blindness and loss of vision.
2. In AIDS patients, the infection may cause endophthalmitis.

Diagnosis:

I. Clinical diagnosis: History and clinical pictures.

II. Laboratory diagnosis:

1. **Wet mount preparation:** Specimens from corneal scraping are examined microscopically to detect trophozoites and cysts.
2. **Giemsa stain** is used to stain the organism for identification.
3. **Stain corneal scraping** with calcofluor white (stain cyst wall) and examined with fluorescent microscopy.
4. **Culture** of corneal scraping or saline solution of contact lenses on non- nutrient agar plate seeded with *Escherichia coli*.
5. **Molecular diagnosis:** PCR.

Treatment

Acanthamoeba keratitis has responded well to oral itraconazole and topical miconazole followed by keratoplasty.

Prevention and control:

The amoebic keratitis, caused by the use of contact lens can be prevented by:

1. Proper cleaning of contact lenses by using sterile saline solution. Avoid using homemade sodium chloride solutions.
2. Avoid using contact lenses while swimming.

Myiasis

***Definition:** It is the invasion of tissues of man or animals by the larval stages of some dipterous flies.

Ocular and nasopharyngeal myiasis

Larvae of flies may find their way to the conjunctiva, ears or the nose. They may cause serious damage destroying the eye, the orbit or the sinuses and may reach the brain with fatal results in extreme cases (e.g. larvae of Wohlfahrtia, Sarcophaga, Oestrus).

Aural myiasis:

Purulent or serous discharge from ears may attract flies to lay eggs or larvae there. Larvae may reach middle or inner ear or even mastoid sinuses and the brain tissues (e.g. Chrysomyia, Wohlfahrtia, Sarcophaga).